Class Ostracoda

Mussel shrimps

Occurrence 5,400 spp. worldwide; in marine, brackish water, freshwater bodies, a few species are terrestrial

Mussel shrimps, also known as seed shrimps, are small, gray, brown, or greenish crustaceans that can measure less than 1/32 in (1 mm) to just over 1¼ in (3 cm) in length. They make up 61 families in 6 orders. Their indistinctly segmented body is fully enclosed in a hinged carapace, which can be variously shaped: circular, oval, or rectangular-oval. Its surface may be smooth or sculptured depending upon the species. Mussel shrimps resemble bivalves (see p.595) in having a carapace with 2 halves, strengthened by calcium carbonate, that can be closed by a muscle. The head is the largest part of the body. All species have a single simple eye, the nauplius eye, in the middle of the head; some also have a pair of compound eyes. Most species have separate sexes, which mate in order to reproduce; however, some are parthenogenetic. Some mussel shrimps emit light to attract mates, and the males of some even synchronize their flashing. Eggs are released into water or stuck to plants; they may be very tough and resistant to drying. Mussel shrimps are very important in aquatic food chains and are found worldwide, from shallow to very deep water, as crawlers, burrowers, or free swimmers. They may scavenge, or feed on detritus or particles suspended in the water; some are predatory. There are more than 10,000 ostracod fossil species, and they are important in oil prospecting since different types of fossils indicate different rock strata.

CYPRIDID MUSSEL SHRIMPS
Found in lakes, pools, and swamps, cypridid mussel shrimps (family Cyprididae) include the majority of freshwater species. Some swim, while others crawl on the bottom. They often reproduce parthenogenetically, and in some species males have never been found. Their eggs are resistant to drying.

CYPRIDINID MUSSEL SHRIMPS
The largest mussel shrimps, including the deep-sea genus Gigantocypris, *belong to this widespread, marine group (family Cypridinidae). Cypridinids are good swimmers. They propel themselves with their second antennae, which project from a small notch in the front end of their carapace valves.*

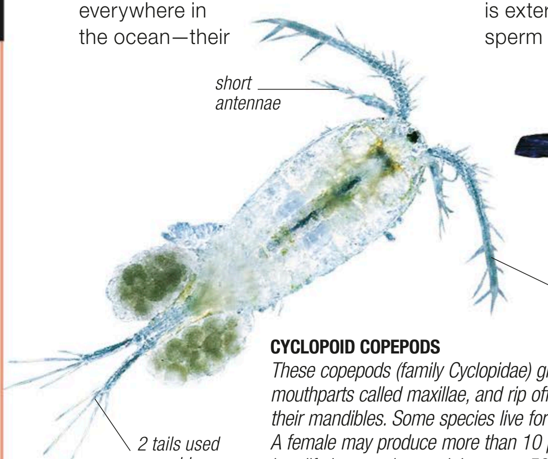
Class Maxillopoda

Maxillopods

Occurrence 18,000 spp. worldwide; in oceans (sediments to deep-sea trenches), freshwater, hot springs, on land

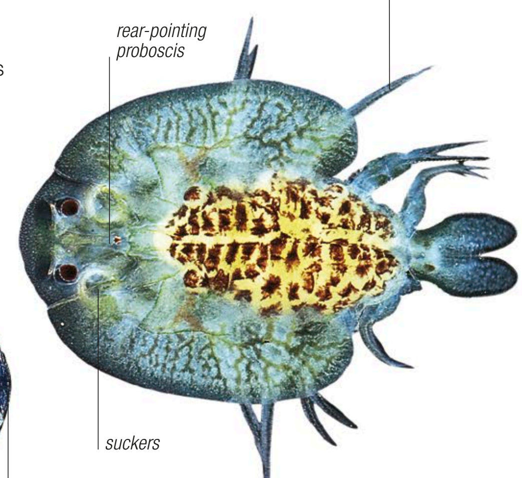
Copepods (Copepoda) and barnacles (Cirripedia) form the most significant subclasses of this large and diverse group of 25 orders and about 310 families. Copepods are abundant everywhere in the ocean—their combined mass exceeds billions of tons, and they are a major source of food for fishes. They are also found in freshwater, from mountain lakes to hot springs. Usually under 3/16 in (5 mm) in length, some parasitic species may grow up to 4 in (10 cm). Generally pale, they have no carapace and lack compound eyes. The head is fused with the trunk, and the first pair of thoracic appendages is used for feeding, while the rest help in swimming. Fertilization is external; the male simply attaches a sperm packet to the female's genitals.

Barnacles are entirely marine crustaceans—they may be stalkless, stalked, or parasitic. They have a shell made up of calcareous plates that surround the body, and 6 pairs of thoracic appendages. Most species are hermaphrodite but some are dioecious, with sexes separated in individuals. Barnacles may grow in such numbers on the underside of a ship, that they reduce its speed by about one third.

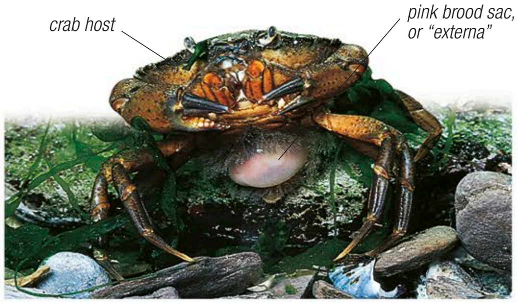


CYCLOPOID COPEPODS
These copepods (family Cyclopidae) grasp prey with mouthparts called maxillae, and rip off small pieces using their mandibles. Some species live for more than 9 months. A female may produce more than 10 pairs of egg sacs in a lifetime, each containing up to 50 eggs. Shown above is a member of the freshwater genus Cyclops.

CALIGID COPEPODS
Members of the family Caligidae are ectoparasites on the bodies of marine and freshwater fishes. They have an enlarged head, with modified appendages to grip the host, and a reduced abdominal region. Some are red due to the presence of hemoglobin in their body. Caligus rapax *(shown here) pierces and sucks blood from the skin or gills of the host.*

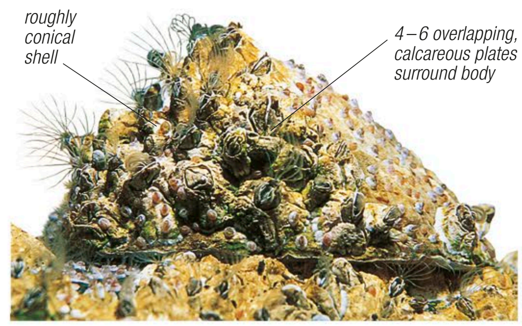


FISH LICE
Belonging to the subclass Branchiura, members of the family Argulidae are commonly called fish lice; their mouthparts are modified for sucking, enabling these parasites to feed on marine or freshwater fishes. A fish infested by them may get fungal infections or even die. Fish lice have a cephalothorax (the head fused with the first thoracic segment), a 3-segmented thorax, and a 2-lobed abdomen. The species shown above belongs to Argulus, *the largest genus in the family.*



SACCULINIDS
The pink swelling on the underside of this crab is part of a sacculinid (family Sacculinidae)—a parasitic cirripede. Adults have highly branched bodies that spread through the tissues of their hosts, absorbing nutrients. Some, including the species of Sacculina *pictured, can infest half the host population in an area.*

COMMON GOOSE BARNACLES
Goose barnacles (family Lepadidae) inhabit surface waters from polar to tropical oceans. The body has 2 parts: the flexible stalk and the main body. The larvae are free swimming but adults attach themselves to algae, floating wood, turtles, and ships. The largest barnacle, Lepas anatifera, *shown here, is up to 32 in (80 cm) long.*



COMMON ACORN BARNACLES
Members of the family Archaeobalanidae can be up to 4 in (10 cm) in width and height, although most are much smaller. They encrust surfaces in large numbers and filter passing food particles. Semibalanus balanoides *(pictured) withstands freezing in Arctic tidal zones, and can survive out of water for up to 9 hours in summer.*